ChatGPT - Lecture Scribes

L6_S1_A.pdf
PDF

Provide me a lecture scribe for the given lecture. The lecture scribe should be exam-oriented, covering all the details properly. It should only contain the given context, i.e. the contents in the given file. Ensure to include the following:
all the definitions, notations, assumptions, conditions that are given.
theorems, axioms, result statements, proofs and derivations. Each of them should be step by step.
When including the given details, ensure to provide the derivations and proofs of only those things that are covered in the pdf. Also include the worked up examples given in the PDF.
Note: there should be no content other than the things covered in the given PDF. make sure to not to introduce any new explanations, examples, intuitions, examples, or even a single thing beyond the provided PDF and context.
Make sure to give the result in a properly mannered and structured way, there should be not random things popping up here and there. Make proper sections, subsections and everything should be properly structured in a single shot. Do not include any extra for feedback or "Here is your....", provide only and only the lecture scribe material.

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

(January 22, 2025)

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1. Random Variables

1.1 Motivation and Concept

A **random variable (RV)** is a function that maps outcomes of a random experiment (sample space) to real numbers.

- The **distribution of a random variable** can be visualized as a bar diagram.
- The **x-axis** represents the values that the random variable can take.
- The **height of the bar at a value** a_i is the probability $\Pr[X = a_i]$.
- Each probability is computed from the probability of the corresponding event in the sample space.

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2. Types of Random Variables

2.1 Discrete Random Variable

A random variable is **discrete** if:

- It has **countable support**.
- It is described using a **Probability Mass Function (PMF)**.
- Probabilities are assigned to **single values**.
- Each possible value has **strictly positive probability**.

2.2 Continuous Random Variable

A random variable is **continuous** if:

- It has **uncountable support**.
- It is described using a **Probability Density Function (PDF)**.
- Probabilities are assigned to **intervals of values**.
- Each individual value has **zero probability**.

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3. Examples of Random Variables

Example 1: Tossing Three Fair Coins

Experiment: Toss 3 fair coins.
Let Y denote the **number of heads** that appear.

- Possible values of Y : $\{0, 1, 2, 3\}$

Probability Calculations

- $\Pr(Y = 0) = \Pr(t, t, t) = \frac{1}{8}$
- $\Pr(Y = 1) = \Pr(t, t, h), \Pr(t, h, t), \Pr(h, t, t) = \frac{3}{8}$
- $\Pr(Y = 2) = \Pr(t, h, h), \Pr(h, t, h), \Pr(h, h, t) = \frac{3}{8}$
- $\Pr(Y = 3) = \Pr(h, h, h) = \frac{1}{8}$

Thus,

$$\Pr(Y = 0) = \frac{1}{8}, \quad \Pr(Y = 1) = \frac{3}{8}, \quad \Pr(Y = 2) = \frac{3}{8}, \quad \Pr(Y = 3) = \frac{1}{8}$$

Since Y must take one of these values:

$$1 = \Pr\left(\bigcup_{i=0}^{3}(Y = i)\right) = \sum_{i=0}^{3} \Pr(Y = i)$$

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4. Probability Mass Function (PMF)

4.1 Definition

A random variable that can take on **at most a countable number of possible values** is said to be **discrete**.

Let X be a discrete random variable with range (possible values)

$$R_X = \{x_1, x_2, x_3, \ldots\}$$

(where the set is **finite or countably infinite**).

The function

$$P_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \ldots$$

is called the **Probability Mass Function (PMF)** of X.

4.2 PMF Property

Since X must take **one of the values** x_k:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

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5. PMF – Worked Example

Example

The probability mass function of a random variable X is given by:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

Step 1: Find the Constant c
Using the PMF condition:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Since,

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

we get:

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}$$

Step 2: Find $\Pr[X = 0]$

$$\Pr[X = 0] = c\frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Find $\Pr[X > 2]$

$$\Pr[X > 2] = 1 - \Pr[X \leqslant 2]$$

$$= 1 - \{\Pr[X = 0] + \Pr[X = 1] + \Pr[X = 2]\}$$

$$= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda})$$

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6. Bayes' Theorem

6.1 Recap

Using conditional probability:

$$\Pr(A \cap B_i) = \Pr(B_i \mid A)\Pr(A)$$

We obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j)\Pr(B_j)}$$

This is known as **Bayes' Formula (Proposition 3.1)**.

6.2 Terminology

- $\Pr(B_i)$: **A priori probability**
  (formed from presupposed or self-evident models)
- $\Pr(B_i \mid A)$: **Posterior probability**
  (computed after observing event A)

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7. Bayes' Theorem – Example 1

Auditorium with 30 Rows

- The auditorium has **30 rows**.
- Row 1 has **11 seats**, Row 2 has **12 seats**, Row 3 has **13 seats**, ..., Row 30 has **40 seats**.
- A door prize is given by:
  1. Randomly selecting a **row** (each row equally likely).
  2. Randomly selecting a **seat within that row** (each seat equally likely).

Tasks

1. Compute the probability that **Seat 15** was selected given that **Row 20** was selected.
2. Compute the probability that **Row 20** was selected given that **Seat 15** was selected.

(Computation is carried out using Bayes' formula as shown in the lecture.)

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8. Bayes' Theorem – Example 2

Communication System (Receiver – RX)

- A communication system sends **binary data**: 0 or 1.
- The receiver (RX) may make errors:
  - A 0 may be detected as 1.
  - A 1 may be detected as 0.
- The system is described using a given **set of conditional probabilities**.

(The posterior probabilities are computed using Bayes' theorem based on the provided conditional probabilities.)